

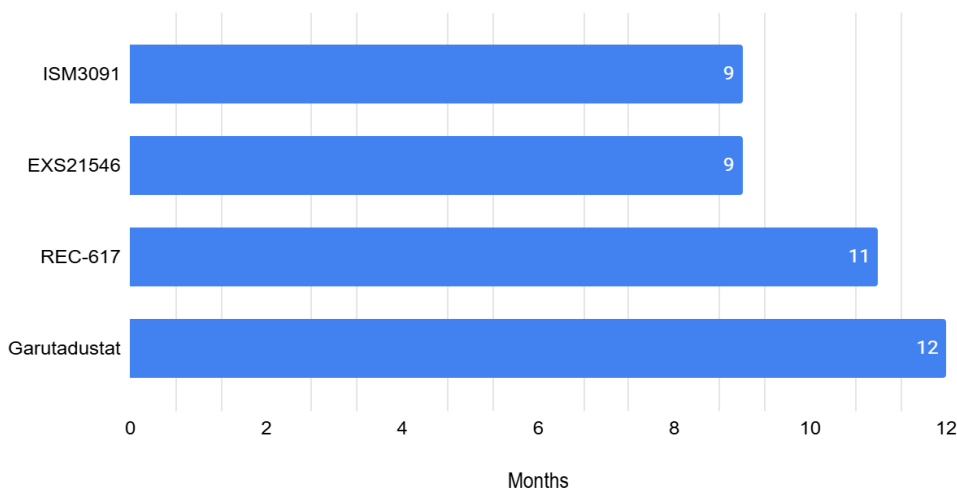
Can AI Actually Design Better Drugs?

Evaluating whether AI is improving drug discovery speed, clinical translation, and commercial value

Developing a new drug takes years of research and testing, and many candidates fail before reaching patients. AI is increasingly being used to identify drug targets, design molecules, screen compounds, and prioritize candidates for further testing.

AI-enabled drug discovery has now reached human clinical trials, while major pharmaceutical companies are partnering with AI-focused biotechnology firms. Selected programs also report much shorter early discovery timelines.[1][2] Still, faster discovery does not guarantee clinical success, and evidence for improved late-stage outcomes remains limited.[3]

Reported Discovery Timelines for Selected AI-Enabled Candidates



Note: Timelines are company-reported and use slightly different discovery milestones. Approximate and "less than" values are plotted using the stated numerical value, so these examples should not be treated as an industry-wide average.[1][2]

What the Evidence Suggests

Faster Early Discovery: Selected programs reached candidate selection in roughly 9 to 12 months, suggesting shorter early discovery timelines.

Human Testing Remains the Hard Part: AI-enabled drugs still need to demonstrate safety and efficacy in clinical trials.

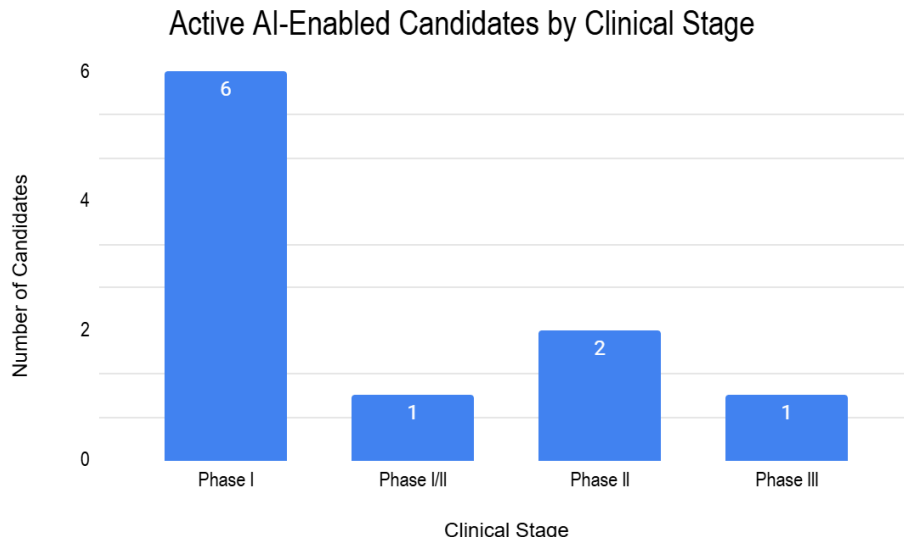
Late-Stage Evidence Is Limited: Few AI-enabled candidates have reached late-stage development, so it is too early to determine whether AI improves overall approval success.

From Algorithms to Human Trials

Clinical progress shows promise, but most AI-enabled candidates remain early in development

Our dataset tracks 15 selected AI-enabled drug candidates across several companies and therapeutic areas. Of the 10 programs still active, most remain in early clinical development. Six are in Phase I, while only one has reached Phase III.

The dataset also includes five discontinued programs. Several progressed beyond initial safety testing before development stopped, showing that AI can help produce candidates suitable for human trials without eliminating the challenges of proving meaningful clinical benefit.



Source: Original analysis of 15 selected AI-enabled clinical programs using clinical trial records, company filings, and pipeline disclosures.

5 of 15 selected candidates were discontinued
3 reached Phase II
1 reached Phase I/II
1 reached Phase I

What This Tells Us

Clinical translation is already happening: AI-enabled candidates in our sample have progressed across multiple stages of human testing.

Late-stage validation remains rare: Only one active candidate in our selected sample has reached Phase III.

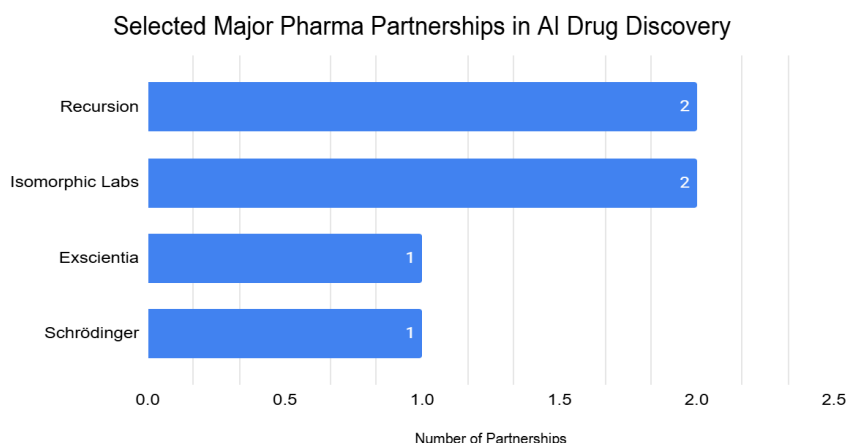
AI does not remove clinical risk: Four of the five discontinued programs progressed beyond Phase I, showing that early clinical progress does not guarantee later success, reinforcing the gap between finding a promising molecule and proving that it works in patients.

Is There a Business Case for AI Drug Discovery?

Major pharmaceutical partnerships show commercial interest even before clinical success is fully proven

AI drug discovery is becoming a business model as well as a research tool. Companies such as Recursion, Isomorphic Labs, Exscientia, and Schrödinger have formed partnerships with major pharmaceutical firms, giving drugmakers access to specialized AI platforms, datasets, and computational expertise.[4][5][7][9]

For AI-focused companies, these partnerships create value through research funding, upfront payments, development milestones, and potential royalties.[4][6][7] This means an AI platform can generate commercial value before one of its drugs reaches the market. Large partnership announcements still require caution, since much of their potential value depends on future clinical and regulatory milestones.[8]



Selected examples from publicly announced pharmaceutical collaborations. Counts reflect selected partnerships included in this analysis, not each company's total collaborations.[4][5][7][9]

The Bigger Picture

Commercial adoption is already happening: Major pharmaceutical companies are partnering with AI-focused drug discovery firms, showing that the technology has value even before an AI-enabled drug reaches the market.

Clinical evidence is less mature: Most active candidates in our sample remain in early clinical development, while several discontinued programs progressed beyond Phase I before stopping. Partnerships and faster discovery therefore do not guarantee clinical success.

Final assessment: AI is already improving parts of drug discovery, particularly target identification, molecular design, and candidate selection. Its clearest advantage today is faster and more efficient early discovery. Whether those gains ultimately lead to more successful medicines remains an open question.

References

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- [4] Recursion Pharmaceuticals. *Partners*. Details collaborations with Sanofi and Roche/Genentech across oncology, immunology, neuroscience, and gastrointestinal oncology.
- [5] Isomorphic Labs. *Partnerships*. Describes its strategic drug-discovery collaborations with Novartis and Eli Lilly, including the expansion of its Novartis research collaboration.
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